COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 212-Automating Server Management	**SECTION:**	CPE31S2
DATE PERFORMED:	8/29/26	**DATE SUBMITTED:**	8/29/26
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Hands-on Activity 5.1 Implement Ansible roles in playbooks

1. PROCEDURE OUTPUT

What procedures did you perform?

Task 1: Use when command for different distributions

1. In the local machine, make sure you are in the local repository directory (CPE232_yourname). Issue the command git pull. When prompted, enter the correct passphrase or password.

```
(base) nikko@Nikko:~/CPE212_Andoy-$ git pull
Already up to date.
(base) nikko@Nikko:~/CPE212_Andoy-$
```

Describe what happened when you issue this command. Did something happen? Why?

- I already pulled the recent changes in my repository last time. That is why it says that the local repository is already up to date.

2. Edit the inventory file and add the IP address of the Centos VM. Issue the command we used to execute the playbook (the one we used in the last activity): ansible-playbook --ask-become-pass install_apache.yml. After executing this command, you may notice that it did not become successful in the Centos VM. You can see that the Centos VM has failed=1. Only the two remote servers have been changed. The reason is that Centos VM does not support "apt" as the package manager. The default package manager for Centos is "yum."

4. Edit the install_apache.yml file and insert the lines shown below

```
(base) nikko@Nikko:~/CPE212_Andoy-/ansible/hoa4-yarn$ ansible-playbook -bK install_apache.yml --limit
centos
BECOME password:

PLAY [all] *********************************************************************************************

TASK [Gathering Facts] *********************************************************************************
[WARNING]: Host 'centos1' is using the discovered Python interpreter at '/usr/bin/python3.9', but futu
re installation of another Python interpreter could cause a different interpreter to be discovered. Se
e https://docs.ansible.com/ansible-core/2.19/reference_appendices/interpreter_discovery.html for more
information.
ok: [centos1]

TASK [update repository index] *************************************************************************
skipping: [centos1]

TASK [install apache2 package] *************************************************************************
skipping: [centos1]

TASK [Add PHP support for apache2] *********************************************************************
skipping: [centos1]

TASK [update repository index] *************************************************************************
ok: [centos1]

TASK [install apache package] **************************************************************************
changed: [centos1]

TASK [Add PHP support for apache] **********************************************************************
changed: [centos1]

PLAY RECAP *********************************************************************************************
centos1                    : ok=4    changed=2    unreachable=0    failed=0    skipped=3    rescued=0
   ignored=0

(base) nikko@Nikko:~/CPE212_Andoy-/ansible/hoa4-yarn$
```

Run ansible-playbook --ask-become-pass install_apache.yml and describe the result.

- Basically the installation was successful and the when condition worked. But I noticed something the value of ansible_distribution isn't referenced into the inventory file meaning I really have to type the CentOS not the one in my inventory.

5. To verify the installations, go to CentOS VM and type its IP address on the browser. Was it successful? The answer is no. It's because the httpd service or the Apache HTTP server in the CentOS is not yet active. Thus, you need to activate it first.

5.1 To activate, go to the CentOS VM terminal and enter the following:

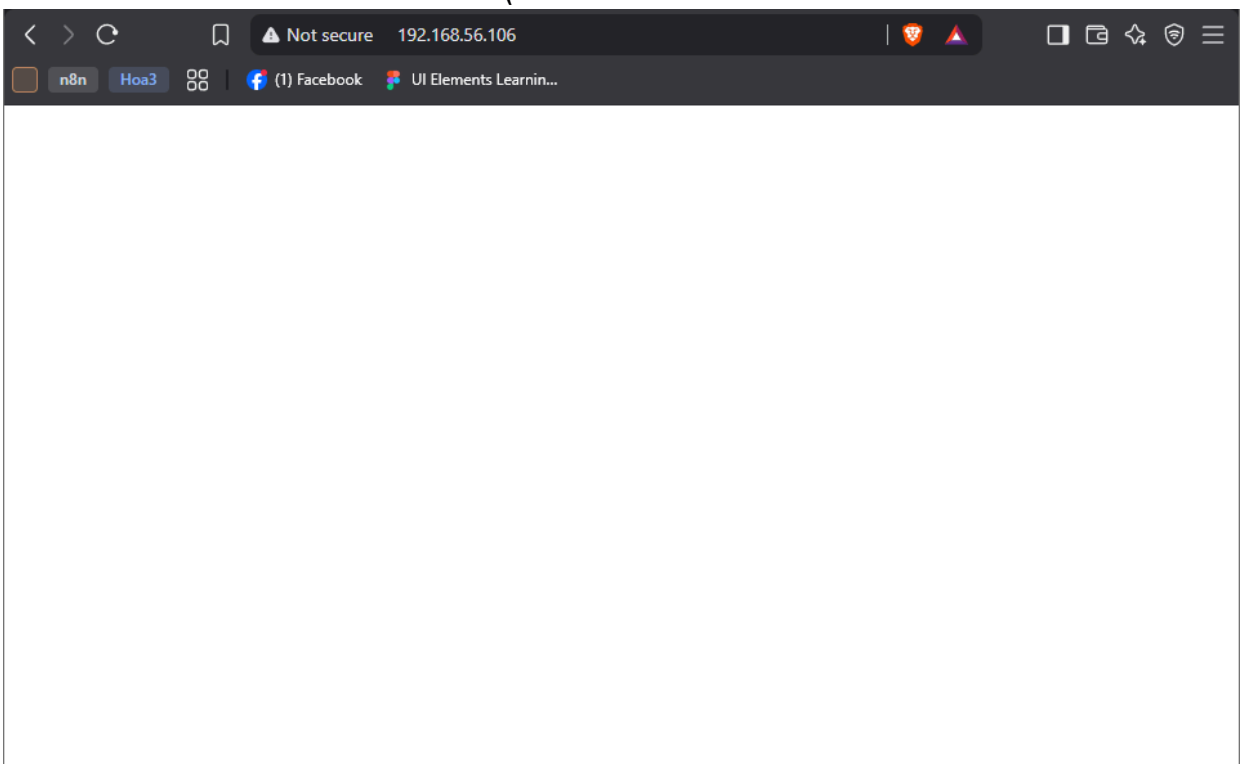
systemctl status httpd

```
[cent-server@localhost ~]$ sudo systemctl start httpd
[sudo] password for cent-server:
[cent-server@localhost ~]$
```

The result of this command tells you that the service is inactive.
5.2 Issue the following command to start the service:
sudo systemctl start httpd

```
[cent-server@localhost ~]$ sudo systemctl start httpd
[sudo] password for cent-server:
[cent-server@localhost ~]$
```

(When prompted, enter the sudo password)
sudo firewall-cmd --add-port=80/tcp
(The result should be a success)sudo s
5.3 To verify the service is already running, go to CentOS VM and type its IP address on the browser. Was it successful? (Screenshot the browser



Yes it was successful but the website doesn't have any contents inside.

Task 2: Refactoring playbook

1. Edit the playbook install_apache.yml. Currently, we have three tasks targeting our Ubuntu machines and 3 tasks targeting our CentOS machine. Right now, we try to consolidate some tasks that are typically the same. For example, we can consolidate two plays that install packages. We can do that by creating a list of installation packages as shown below:

```
(base) nikko@Nikko:~/CPE212_Andoy-/ansible/hoa4-yarn$ ansible-playbook -bK install_apache.yml -l cento
s
BECOME password:

PLAY [all] *********************************************************************************************

TASK [Gathering Facts] *********************************************************************************
[WARNING]: Host 'centos1' is using the discovered Python interpreter at '/usr/bin/python3.9', but futu
re installation of another Python interpreter could cause a different interpreter to be discovered. Se
e https://docs.ansible.com/ansible-core/2.19/reference_appendices/interpreter_discovery.html for more
information.
ok: [centos1]

TASK [update repository index] *************************************************************************
skipping: [centos1]

TASK [Add PHP support for apache2 & install apache2] ***************************************************
skipping: [centos1]

TASK [update repository index] *************************************************************************
ok: [centos1]

TASK [Add PHP support for apache install httpd] ********************************************************
ok: [centos1]

PLAY RECAP *********************************************************************************************
centos1                    : ok=3    changed=0    unreachable=0    failed=0    skipped=2    rescued=0
   ignored=0
```

Run ansible-playbook --ask-become-pass install_apache.yml and describe the result

- The download is successful. Instead of downloading the necessary files 1 by 1 I used the like list type.

3. Finally, we can consolidate these 2 plays in just 1 play. This can be done by declaring variables that will represent the packages that we want to install. Basically, the apache_package and php_package are variables. The names are arbitrary, which means we can choose different names. We also take out the line when: ansible_distribution. Edit the playbook install_apache.yml again and make sure to follow the below image. Make sure to save the file and exit.

```
(base) nikko@Nikko:~/CPE212_Andoy-/ansible/hoa4-yarn$ ansible-playbook -bK install_apache.yml -l cento
s
BECOME password:

PLAY [all] *********************************************************************************************

TASK [Gathering Facts] *********************************************************************************
[WARNING]: Host 'centos1' is using the discovered Python interpreter at '/usr/bin/python3.9', but futu
re installation of another Python interpreter could cause a different interpreter to be discovered. Se
e https://docs.ansible.com/ansible-core/2.19/reference_appendices/interpreter_discovery.html for more
information.
ok: [centos1]

TASK [Add PHP support for apache2 & install apache2] ***************************************************
skipping: [centos1]

TASK [Add PHP support for apache install httpd] ********************************************************
[ERROR]: Task failed: Finalization of task args for 'ansible.builtin.dnf' failed: Error while resolvin
g value for 'name': 'php_package' is undefined
```

- The script failed and the error returned is “Error while resolving value for ‘name’: ‘php_package’ is undefined.
-

4. Unfortunately, task 2.3 was not successful. It’s because we need to change something in the inventory file so that the variables we declared will be in place. Edit the inventory file and follow the below configuration:

```
(base) nikko@Nikko:~/CPE212_Andoy-/ansible/hoa4-yarn$ ansible-playbook -bK install_apache.yml -l cento
s
BECOME password:

PLAY [all] *********************************************************************************************

TASK [Gathering Facts] *********************************************************************************
[WARNING]: Host 'centos1' is using the discovered Python interpreter at '/usr/bin/python3.9', but futu
re installation of another Python interpreter could cause a different interpreter to be discovered. Se
e https://docs.ansible.com/ansible-core/2.19/reference_appendices/interpreter_discovery.html for more
information.
ok: [centos1]

TASK [Add PHP support for apache2 & install apache2] ***************************************************
skipping: [centos1]

TASK [Add PHP support for apache install httpd] ********************************************************
ok: [centos1]

PLAY RECAP *********************************************************************************************
centos1                    : ok=2    changed=0    unreachable=0    failed=0    skipped=1    rescued=0
   ignored=0
```

- Scrit is successful in the side of ubuntu and it skipped te centos because that is no the package of its own.

Finally, we still have one more thing to change in our install_apache.yml file. In task 2.3, you may notice that the package is assign as apt, which will not run in CentOS. Replace the apt with package. Package is a module in ansible that is generic, which is going to use whatever package manager the underlying host or the target server uses. For Ubuntu it will automatically use apt, and for CentOS it will automatically use dnf. Make sure to save the file and exit. For more details about the ansible package, you may refer to this documentation:

```
(base) nikko@Nikko:~/CPE212_Andoy-/ansible/hoa4-yarn$ ansible-playbook -bK install_apache.yml -l cento
s
BECOME password:

PLAY [all] *********************************************************************************************

TASK [Gathering Facts] *********************************************************************************
[WARNING]: Host 'centos1' is using the discovered Python interpreter at '/usr/bin/python3.9', but futu
re installation of another Python interpreter could cause a different interpreter to be discovered. Se
e https://docs.ansible.com/ansible-core/2.19/reference_appendices/interpreter_discovery.html for more
information.
ok: [centos1]

TASK [install apache and php packages] *****************************************************************
ok: [centos1]

PLAY RECAP *********************************************************************************************
centos1                    : ok=2    changed=0    unreachable=0    failed=0    skipped=0    rescued=0
   ignored=0

(base) nikko@Nikko:~/CPE212_Andoy-/ansible/hoa4-yarn$
```

- **This part optimizes the install_apache2 by using the {{}} and putting the necessary variables and defining them into the inventory file.**

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

1. Create a playbook that could do the previous tasks in Red Hat OS.

```
(base) nikko@Nikko:~/CPE212_Andoy-/ansible/hoa4-yarn$ ansible-playbook -bK ../installing_nginx.yml
BECOME password:

PLAY [centos] ******************************************************************************************

TASK [Gathering Facts] *********************************************************************************
[WARNING]: Host 'centos1' is using the discovered Python interpreter at '/usr/bin/python3.9', but futu
re installation of another Python interpreter could cause a different interpreter to be discovered. Se
e https://docs.ansible.com/ansible-core/2.19/reference_appendices/interpreter_discovery.html for more
information.
ok: [centos1]

TASK [Installing nginx] ********************************************************************************
changed: [centos1]

PLAY RECAP *********************************************************************************************
centos1                    : ok=2    changed=1    unreachable=0    failed=0    skipped=0    rescued=0
   ignored=0
```

Source code:

Inventory

```
    centos:
      hosts:
        centos1:
          ansible_host:
192.168.56.106
          ansible_user:
cent-server
          nginx_package: nginx
```

Installing_nginx

```
---
- hosts: centos
  become: true
  tasks:
    - name: Installing nginx
      package:
        name:
          - "{{ nginx_package }}"
        state: latest
        update_cache: yes
```

Reflections:

Answer the following:

1. Why do you think refactoring of playbook codes is important?
   - We refactor the code to ensure that it is optimized and efficient when executed. Also by cleaning up the code we can easily read and identify which part of code is the problem when an error occurred, this also lessen the debugging phase of the code.
2. When do we use the “when” command in playbook?
   - When is used to install different OS just like in the activity because we run different OS in the activity by using when we can ensure the redundancy of the code.

Conclusion:

- **In conclusion this activity helped me to optimize the code and ensure the efficiency of it when running. Using "-" to list the download packages to make the install file clean and optimized and using variables on it for clarification of name. I also observed that I don't have to specifically define which OS will download the package since clearing the package: ensures any kind of OS to installed into. Although, the inventory file is full of code since that is where we define the needed packages and just use the variables in the install file.**

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

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Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use:

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

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